

Lithium Ion Battery

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Outlines

Batteries with nonaqueous electrolyte

➤ **Lithium ion Battery**

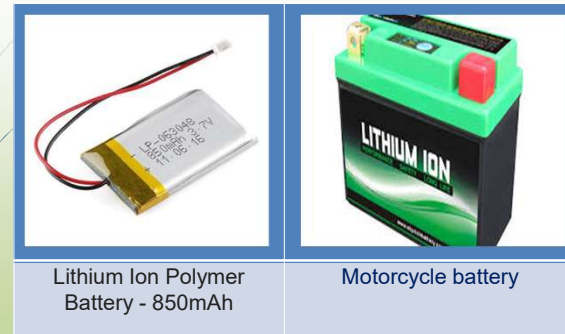
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Lithium ion Battery

Invented in 1980 by American physicist
Professor John Goodenough .

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Li-ion Battery



Lithium Ion Polymer
Battery - 850mAh

Motorcycle battery

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Li-ion Battery

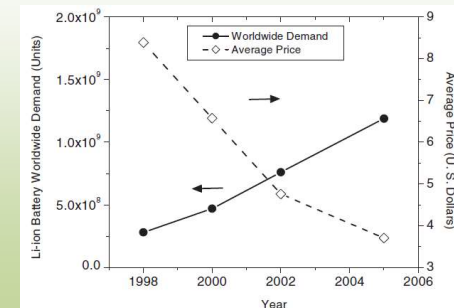


Solar energy

Phone Battery

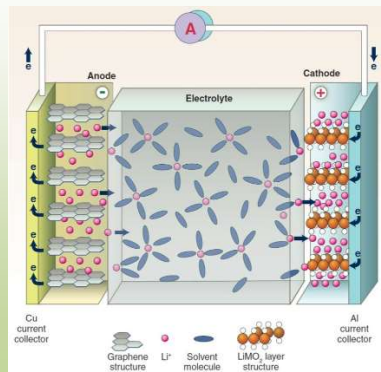
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Current and anticipated worldwide demand and average price for Li-ion batteries.



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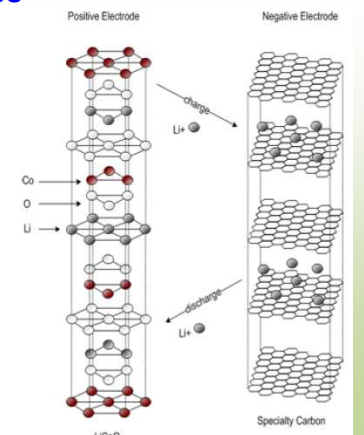
Li-ion Battery: Main Components



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Li-ion batteries

- Positive electrode: Lithiated form of a transition metal oxide (lithium cobalt oxide- LiCoO_2 or lithium manganese oxide LiMn_2O_4)
- Negative electrode: Carbon (C), usually graphite (C_6)
- Electrolyte: solid lithium-salt electrolytes (LiPF_6 , LiBF_4 , or LiClO_4) and organic solvents (ether)



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Requirements for Li-ion : Positive Electrode Materials

- High free energy of reaction with lithium
- Can incorporate large quantities of lithium
- Reversibly incorporates lithium without structural change
- High lithium ion diffusivity
- Good electronic conductivity
- Insoluble in the electrolyte
- Prepared from inexpensive reagents
- Low cost synthesis

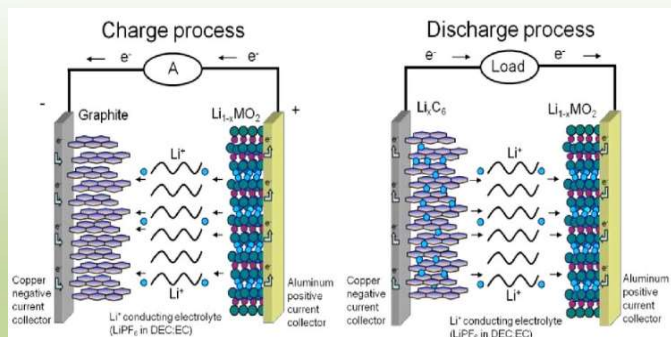
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Requirements for Li-ion separators

- ❑ High machine direction strength to permit automated winding
- ❑ Does not yield or shrink in width
- ❑ Resistant to puncture by electrode materials
- ❑ Effective pore size less than 1 μm
- ❑ Easily wetted by electrolyte
- ❑ Compatible and stable in contact with electrolyte and electrode materials

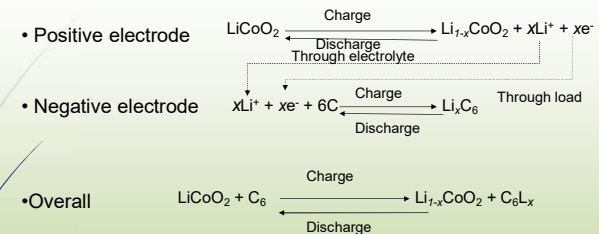
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Li-ion Battery: Working Principle



Li-ion batteries

- Chemical reaction (Charging)



- In the above reaction x can be 1 or 0
- With charge the Co is oxidized from Co^{3+} to Co^{4+} . The reverse process (reduction) occurs when the battery is being charged.

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Advantages of Li-ion Batteries

- Sealed cells; no maintenance required
- Long cycle life
- Broad temperature range of operation
- Long shelf life
- Low self-discharge rate
- Rapid charge capability
- High rate and high power discharge capability
- High coulombic and energy efficiency
- High specific energy and energy density
- No memory effect

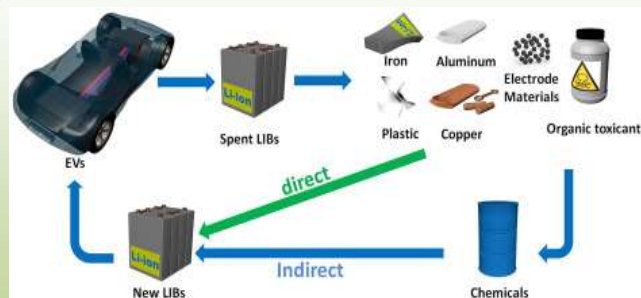
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Disadvantages of Li-ion Batteries

- Moderate initial cost
- Degrades at high temperature
- Need for protective circuitry
- Capacity loss or thermal runaway when overcharged.
- Venting and possible thermal runaway when crushed
- Cylindrical designs typically offer lower power density than NiCd or NiMH

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Recycling of lithium-ion batteries



[Li-ion batteries video](#) 6.35 min [Tesla Motor](#)



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Different SI Units For Batteries

➤ **Ampere (A or amp) :** (unit of [electric current](#))

Refers to the amount of constant current (electric charge) that produces a force equal to 2×10^{-7} newton per meter on two straight parallel conductors.

➤ **Ampere hour (Ah or amp hr):** (unit of [electric charge](#))

A steady flow of 1 ampere for 1 hour (straight) is 1 Ah. The equivalent is 3600 coulomb.

➤ **Coulomb (C):**(unit of [electric charge](#))

Accumulated per second of a 1 amp current is measured as 1 coulomb (C).

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Different SI Units For Batteries

➤ **Farad (F):** (electric capacity)

One farad is the ability of the conductor to store 1 C (coulomb) of charge for every volt of potential difference between the two conductors.

➤ **Faraday (Fd):**(unit of [electric charge](#))

1 Fd is equivalent to the product of a charge on a single [electron](#) and Avogadro's constant.

➤ **Joule (J):** (energy)

A joule is the amount of energy exerted as the force of 1 newton is applied over a displacement of 1 meter. 1 joule is also equal to the energy of 1 watt per second.

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Different SI Units For Batteries

➤ **Ohm (Ω) :** (unit of electrical resistance)

One ohm is the amount of resistance that needs a potential difference of 1 volt per ampere.

➤ **Siemens (S):** (unit of electric conductance)

One siemens is when a conductor carries a current of one ampere per volt of potential.

➤ **Volt (V):** (unit of electrical potential)

A volt represents a potential of one joule per coulomb of charge.

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Different SI Units For Batteries

➤ **Henry (H):** (electric [inductance](#))

1 H usually requires a flux of 1 weber per ampere of the current induced.

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Nickel-cadmium battery

❖ Invented in 1899 by Waldmar Jungner.

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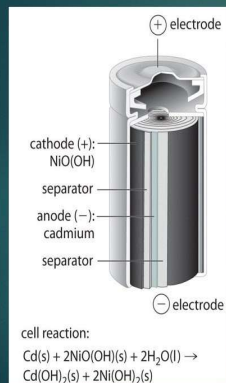
Ni-Cd aircraft battery

Ni-Cd battery

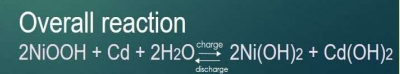
Glory Ni-Cd battery

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1- Ni-Cd (Nickel Cadmium battery)



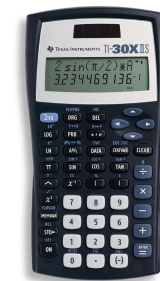
Battery reactions:



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Application of Nickel-cadmium battery

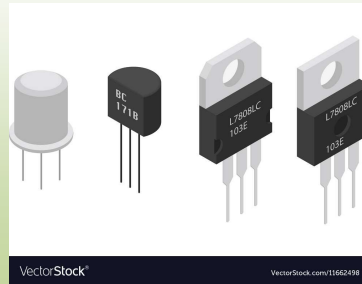
► Calculators



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Application of Nickel-cadmium battery

Transistors



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Application of Nickel-cadmium battery

Cordless appliances



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Application of Nickel-cadmium battery

Electronic Flash Units



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Advantages of Ni-Cd Batteries

- Available in different size and shapes.
- Have ability to deliver full power output until end of cycle
- Very low internal resistance
- Simple and fast recharge over 1000 cycles (if properly maintained)
- Operate in a range of temperatures
- Tolerates abuse well and performs well after long periods of storage

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Disadvantages of Ni-Cd Batteries

- It is three to five times more expensive than lead-acid.
- Self- discharge up to 10% in a day.
- Its materials are toxic and the recycling infrastructure for larger nickel-cadmium batteries is very limited.